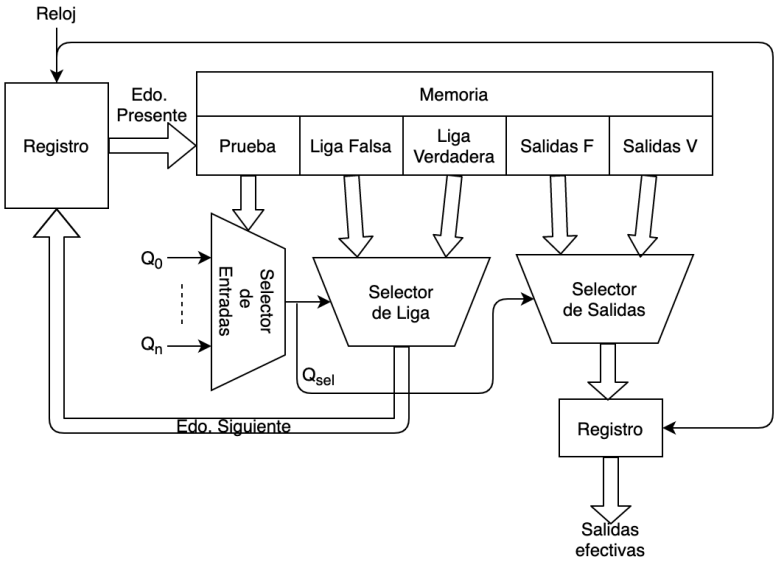
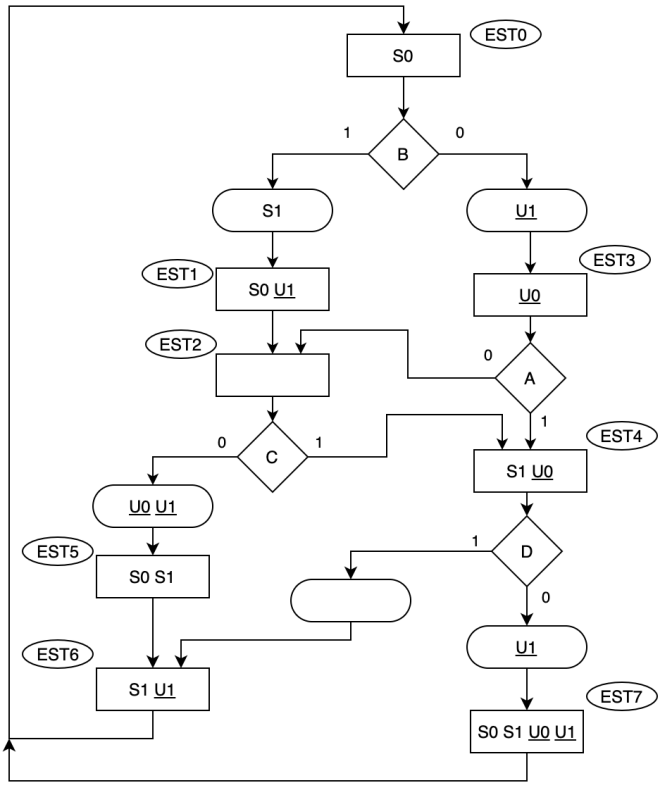
Arquitectura del direccionamiento Entrada – Estado modificada para soportar salidas condicionales:




Ejercicio

1. Obtener el tamaño y el contenido de la memoria para la siguiente carta ASM. Utilizar la arquitectura modificada del direccionamiento entrada estado para realizar el ejercicio.



Tamaño de la memoria

$$n = bits\ los\ estados$$

$$m = bits\ de\ prueba + bits\ Liga\ Falsa + bits\ Liga\ Verdadera + bits\ Salidas\ Falsas + bits\ Salidas\ Verdadderas$$

$$Tamaño\ de\ la\ memoria = 2^n * m$$

Estados: 8 Entradas:4 Salidas: 4

N = 3
M = 2 + 3 + 3 + 4 + 4= 16

$2^3 * 16 = 128\ bits$

Valores de Prueba:

A: 00
B: 01
C: 10
D: 11

| Dirección | | | Contenido | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Edo. Presente | | | Prueba | | L F | | | L V | | | S F | | | | S V | | | |
| P2 | P1 | P0 | K1 | K0 | F2 | F1 | F0 | V2 | V1 | V0 | S1 | S0 | $\overline{U1}$ | $\overline{U0}$ | Z1 | Z0 | $\overline{V1}$ | $\overline{V0}$ |
| 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | * | * | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 1 | 0 | 1 | * | * | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | * | * | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | * | * | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |